Region-Aware MoE Network for Hyperspectral and Multispectral Image Fusion

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***Abstract*—Existing transformer-based hyperspectral image (HSI) and multispectral image (MSI) fusion methods often suffer from limited cross-modal interaction and insufficient capability to model heterogeneous spatial and spectral patterns across different regions, which restricts their ability to fully exploit modality complementarity. To address these issues, we propose a region-aware mixture of experts (MoE) network for HSI and MSI fusion. The proposed framework consists of three key components. First, an attention sharing module (ASM) is designed to explicitly propagate attention information between modalities, enabling more balanced interaction between spatial and spectral representations. Second, the region-aware MoE (RAMoE) is introduced, where multiple region-aware experts (RAEs) capture region-specific feature variations, while an invariant-weight expert (IWE) provides global contextual guidance, improving robustness and generalization. Third, a spatial and spectral reconstruction module (SSRM) is employed to jointly recover spatial details and spectral information through residual learning. Extensive experiments on benchmark datasets demonstrate that the proposed method consistently outperforms state-of-the-art fusion methods in terms of both quantitative metrics and visual quality. Furthermore, experiments on real-world datasets verify the practical effectiveness of RAMoE in recovering fine spatial details while preserving spectral fidelity. The code will be released at https://github.com/XXn25/RAMoE**

***Index Terms*—Attention sharing, hyperspectral image (HSI), image fusion, multispectral image (MSI), region-aware MoE (RAMoE).**

NOMENCLATURE

$\mathbf{H}, \widetilde{\mathbf{H}}$	LR-HSI and bicubic-upsampled HSI.
$\widehat{\mathbf{H}}, \mathbf{H}^*$	Reconstructed and reference HR-HSI.
$\mathbf{M}$	HR-MSI.
X, Y, and b	Height, width, and band number of the HR-MSI.
x, y, and B	Height, width, and band number of the LR-HSI.
$\mathbf{E}_k$ and $\mathbf{W}_k$	Outputs and weight of kth RAE.
$\mathbf{A}$ and $\mathbf{O}$	Outputs of ASM and RAMoE.

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I. INTRODUCTION

HYPERSPECTRAL images contain hundreds of continuous and narrow spectral bands, spanning from visible to near-infrared wavelengths. This equips hyperspectral image (HSI) with significant advantages in accurately identifying the material properties of ground objects, leading to their widespread application in various remote sensing fields such as dynamic environmental monitoring [1] and land cover classification [2], [3]. However, due to hardware limitations and budget constraints of imaging devices, it is challenging to simultaneously acquire HSI with both high spatial and spectral resolution [4], [5]. Specifically, while HSI maintains high spectral resolution, their spatial resolution is often low, which limits their practical value and application scope. In contrast, high-resolution MSI (HR-MSI) has lower spectral resolution but higher spatial resolution. Typically, low-resolution HSI (LR-HSI) and HR-MSI are acquired by different sensors, respectively. LR-HSI and HR-MSI can be regarded as spatially and spectrally degraded versions of an underlying high-resolution HSI (HR-HSI) [6], [7]. The image reconstruction task is inherently ill-posed, and performing super-resolution reconstruction using only LR-HSI is challenging [8], [9]. Introducing HR-MSI as auxiliary information can significantly improve reconstruction quality [10], [11]. Therefore, designing efficient and high-performance fusion algorithms that fully exploit the complementary spatial and spectral information from multiple data sources to reconstruct HR-HSI has become a current research hotspot in the field of remote sensing image processing.

With the development of deep learning, data-driven end-to-end fusion methods have gradually become dominant. These methods can automatically learn implicit prior constraints from massive data and often demonstrate excellent performance [12], [13]. Deep learning-based approaches for fusing LR-HSI and HR-MSI have become the predominant research direction in recent years. These fusion methods generally fall into

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two main categories: convolutional neural network (CNN)-based models that capture local spatial–spectral correlations [14], [15], [16] and transformer-based models that exploit global dependencies [17], [18], [19]. CNN-based methods, through designing multilayer convolutional structures, effectively extract spatial and spectral features of images and learn the optimal mapping between observed and target images, achieving remarkable fusion results. However, limited by the local receptive field of CNN, they struggle to fully capture global long-range dependencies in images, especially when processing large-scale and multiscale information. To address this issue and obtain richer feature representations, several CNN-based methods have integrated multiscale feature extraction and fusion mechanisms into LR-HSI and HR-MSI fusion frameworks [20], [21], [22]. Nevertheless, due to the intrinsic locality of convolution operations, which are primarily effective at extracting local spatial patterns, CNN-based methods still struggle to fully leverage the rich and high-dimensional spectral information of HSI, and this may lead to oversmoothing of spectral curves and result in distortion [23], [24]. On the other hand, CNNs can only perceive information within a limited neighborhood. Although stacking convolutional layers can indirectly expand the receptive field, modeling global spatial–spectral contextual relationships remains challenging.

Networks based on transformers have gradually been introduced into the field of remote sensing image fusion in recent years due to their excellent capability of modeling global information. Through the self-attention mechanism, they can capture long-range dependencies, making them suitable for handling the rich spectral correlations in HSI, as well as the spatial details in HR-MSI [25], [26]. While the self-attention mechanism in the transformer is effective for extracting features within a single image or modality, it is confined to a single modality, limiting its capacity to capture interactions across multiple modalities. However, in the fusion of LR-HSI and HR-MSI, it is essential to capture mutual attention between the two modalities. The self-attention mechanisms are inherently limited in this regard, as they cannot effectively model the cross-modal relationships required for optimal fusion. This limitation may result in insufficient information exchange between the LR-HSI and HR-MSI modalities, thereby restricting the network's ability to fully exploit the complementary spectral and spatial features of the two inputs and potentially impairing the overall performance and accuracy of the fusion process. More importantly, insufficient cross-modal interaction may negatively affect downstream remote sensing applications that rely on both spatial detail and spectral fidelity. For example, inaccurate fusion may introduce spectral distortions that degrade land cover classification performance [27], [28], especially for spectrally similar materials. In addition, the loss of fine spatial structures may obscure small-scale objects or boundaries, reducing the reliability of change detection [29]. These issues highlight that effective cross-modal interaction is not only crucial for improving reconstruction quality but also directly impacts the practical applicability of fused HSI products in real-world remote sensing tasks.

To address this issue, Ma et al. [30] proposed the cross-attention mechanism that extends the conventional self-attention from a single-modality operation to a multimodality one. By allowing the attention computation to incorporate features from both LR-HSI and HR-MSI, the resulting output is no longer solely dependent on the features of a single modality, enabling more effective integration of complementary spectral and spatial information. Sun et al. [31] proposed the multiscale deep cross-fusion transformer (MDC-FusFormer), which improves the attention mechanism by generating attention maps from the original low-resolution images. This design enables effective multimodal attention interaction between LR-HSI and HR-MSI features, thereby enhancing spatial–spectral feature fusion and improving overall reconstruction quality.

Although these cross-attention-based methods have improved the traditional transformer architecture to enable a certain degree of cross-modal information exchange, they still fail to realize true modality-level interaction between LR-HSI and HR-MSI within the attention mechanism. In existing designs, each modality computes its own attention map independently, without allowing one modality's attention to be leveraged by the other. As a result, the complementary spectral and spatial information across modalities cannot be fully exploited, limiting the effectiveness of feature fusion and the overall performance of the model. In addition, the single-expert-like structure in the feed-forward network (FFN) of the transformer faces limitations in representation capacity and generalization ability, making it difficult to effectively adapt to diverse and complex input features. To solve this problem, some researchers have proposed mixture of experts (MoE) networks [32], [33], and different experts can focus on different types of input patterns or features, thus achieving a division of labor and collaboration among experts. This significantly enhances the model's expressiveness and generalization performance on complex data, leading to more robust results.

To address the aforementioned challenges, we propose the region-aware MoE (RAMoE) network for LR-HSI and HR-MSI fusion. The network leverages an attention sharing module (ASM) to enable multimodal interaction at the attention level, jointly utilizing information from both LR-HSI and HR-MSI to fully exploit their complementary spectral and spatial characteristics. RAMoE integrates outputs from RAEs, each adaptively focusing on the regions it is most interested in, through a weighted fusion mechanism. In addition, the spatial and spectral reconstruction module (SSRM) extracts original spectral and spatial features from LR-HSI and HR-MSI, further enhancing overall performance. Extensive experiments on both simulated and real-world remote sensing images demonstrate that RAMoE surpasses existing state-of-the-art methods, achieving superior super-resolution performance. In summary, this work makes three primary contributions.

1) To fully leverage the attention information from both LR-HSI and HR-MSI, we propose the ASM that enables joint utilization of features from different modalities during the attention computation.
2) We propose the RAMoE that adaptively fuses multiple RAEs using pixelwise gating weights. In addition,

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the invariant-weight expert (IWE) with spatially fixed weights is integrated into the RAMoE, providing globally consistent guidance that complements the spatially adaptive experts.

3) To fully exploit information from LR-HSI and HR-MSI, we propose SSRM to extract spectral and spatial features from the original input images. By preserving the initial image details, SSRM enhances reconstruction quality and significantly improves the overall performance of the network.

The rest of this article is summarized as follows. Section II discusses some deep learning-based methods related to LR-HSI and HR-MSI fusion. Section III details the structure of the RAMoE network. Section IV presents the experimental results and analysis of the proposed method. Finally, the work of this article is summarized in Section V.

II. Related Works

In this section, we introduce the research progress of LR-HSI and HR-MSI fusion, including recent deep learning methods based on CNNs and transformers.

A. CNN Methods in LR-HSI and HR-MSI Fusion

Unlike the traditional methods that require manual design of prior information [34], [35], deep learning-based methods leverage large datasets to learn data-driven priors, and these algorithms have exhibited significant potential and advantages in LR-HSI and HR-MSI fusion tasks, primarily due to their strong capability in feature extraction. In recent years, CNNs have demonstrated impressive performance in addressing LR-HSI and HR-MSI fusion problems. For instance, Dian et al. [36] initialize the HR-HSI by solving a Sylvester equation and then use a deep residual CNN to learn the priors of the HR-HSI. Xie et al. [37] exploited the domain knowledge encoded into the HSI observation model. They jointly exploit the observation model and the low-rank constraints of the spectral to reconstruct HR-HSIs. In order to achieve information fusion between spatial modes and spectral modes, Zhang et al. [38] proposed an interpretable spatial–spectral reconstruction network (SSRNet) that introduces spatial edge loss and spectral edge loss to constrain spatial variations and spectral distortions during the image fusion process. To enable fusion in the feature domain and reconstruct the image from the fused features, Liu et al. [39] encoded the spatial and spectral features independently by a two-stream fusion network (TFNet) and then decoded the HR-HSI using the features of spatial and spectral. ResTFNet [39] was proposed by adding more skip-connection operations on the TFNet, which achieved better reconstruction results. Dong et al. [40] integrated imaging models within the network architecture, proposing the model-guided deep convolutional network (MoG-DCN). They use a proximal gradient descent (PGD)-based unfolding network, incorporating a U-Net [41] denoiser for prior learning, enabling effective end-to-end optimization in LR-HSI and HR-MSI fusion. CNN-based methods have greatly improved LR-HSI and HR-MSI fusion by learning powerful spatial–spectral representations. However, these approaches effectively capture local spatial–spectral features, and their inherent locality constrains the modeling of global context, which is crucial for accurate fusion. This limitation has driven recent research toward transformer-based methods that naturally model long-range dependencies.

B. Transformer Methods in LR-HSI and HR-MSI Fusion

The successful deployment of transformers in image processing has established them as a powerful backbone for a wide range of vision tasks. Building upon this foundation, transformers have recently been applied to LR-HSI and HR-MSI fusion, demonstrating notable effectiveness. Attention mechanism in the transformer can capture long-range dependencies across both spatial and spectral dimensions, facilitating the efficient integration of complementary information from multiple modalities. Hu et al. [42] designed a transformer-based architecture for the LR-HSI and HR-MSI fusion, which can globally explore the intrinsic relationship within features and reduce the burden of training the whole data in a smaller mapping space by the spatial residual. Sun et al. [43] proposed a dual spatial–spectral pyramid network (DSPNet) to achieve comprehensive integration of spatial and spectral information. DSPNet leverages self-attention and cross-attention mechanisms at multiple scales to effectively fuse spatial and spectral features. Ma et al. [44] addressed the limitation of 2D-CNN-based methods in capturing spectral characteristics by proposing a hybrid approach. Specifically, they employed a 3D-CNN to construct a spectral prior learning network and utilized Swin-transformer blocks to design a spatial prior learning network. These two networks are jointly integrated to reconstruct the HR-HSI, effectively leveraging both spectral and spatial priors. To address the limitation of the transformer in fully integrating and coordinating information from the two modalities, Ma et al. [30] proposed the dual cross transformer (DCTransformer) for LR-HSI and HR-MSI fusion, building upon the Swin-transformer architecture. In this design, attention matrices are generated using both LR-HSI and HR-MSI within the attention module, while modality-specific query matrices are employed during computation to effectively achieve two modalities fusion. To flexibly accommodate the varying super-resolution requirements in real-world scenarios, Wang et al. [45] proposed a resolution-independent unfolding algorithm and a multiscale training framework, termed the perceptive spectral transformer unfolding network (PSTUN). This approach leverages real LR-HSI and HR-MSI as priors to reconstruct the spectral and spatial degradation processes, enabling adaptive transfer of spectral information and, thereby, enhancing fusion accuracy. Overall, transformer-based methods typically either extract spatial and spectral features before fusion or introduce cross-attention modules to merge information from different modalities. However, extracting spatial and spectral features separately may lead to insufficient feature interaction, limiting the complementary benefits between modalities. On the other hand, while cross-attention mechanisms incorporate features from one modality into the attention computation of another, they still do not realize attention-level sharing across modalities. To fully leverage information from different modalities and

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enable attention-level interaction, we propose the ASM that shares attention information between modalities and enhances intermodal feature interaction.

C. Mixture of Experts

The MoE framework, originally developed in the context of large language models (LLMs) to increase model capacity [46], is an effective strategy for leveraging multiple specialized networks, or experts, to handle different aspects of complex data. In this approach, each expert is responsible for processing a specific subset of the input, while a gating mechanism adaptively combines their outputs to form the final result. MoE enables the model to capture diverse feature patterns, focus on complementary regions, and enhance overall capacity without a proportional increase in computational cost. Moreover, MoE has demonstrated strong adaptability and interpretability in low-level vision tasks, including image enhancement [47] and image classification [48].

For instance, to simultaneously leverage both local and global information, Cao et al. [49] proposed a framework in which the mixture of local experts (MoLE) captures adaptive local features, while the mixture of global experts (MoGE) extracts complementary global texture and contrast, and their fusion enhances overall image quality. To enable MoE to adaptively adjust the number of experts based on task difficulty, Chen et al. [50] adjust the number of experts according to the difficulty of each visual task, effectively avoiding overfitting of simple tasks or the underfitting of challenging scenarios. In the context of LR-HSI and HR-MSI fusion, different experts of MoE can concentrate on distinct spectral or spatial features, and the weighted combination of their outputs ensures that the most informative information is prioritized, resulting in higher-quality fused images. He et al. [51] proposed the attribute-aware mixture of fusion experts (AMoFE), which employs multiple experts to extract spatial, spectral, and edge information from the image and combines their outputs through a weighted fusion to produce the final result.

In these MoE-based fusion methods, two main strategies are commonly adopted. One aims to increase network capacity by introducing multiple experts, where a gating network selectively activates or suppresses expert outputs. The other focuses on extracting diverse features using different experts and integrating them through weighted fusion. Depending on the expert design and feature emphasis, the resulting representations may differ in their feature distributions, which can make the fusion process more challenging in certain scenarios.

Inspired by these observations, we propose the RAMoE network that places additional emphasis on the consistency and coordination of expert feature representations during fusion. Unlike conventional MoE approaches that separate experts into those capturing different feature categories [51] or into global and local experts [49], each of the RAEs in RAMoE adaptively focuses on the regions it is most interested in. The RAEs in RAMoE are used for weighted fusion, and the fusion process emphasizes the regions that each expert adaptively attends to, producing more accurate and coherent results. All RAEs employ the same architecture and input feature space, which helps maintain a consistent embedding space across expert outputs and facilitates subsequent feature integration. Furthermore, the gating network produces pixelwise weight maps that match the spatial and channel dimensions of the expert feature outputs. This design not only allows the fusion process to flexibly adapt to local spatial and spectral variations but also helps ensure compatibility among expert features by guiding their contributions in a coherent, region-adaptive manner, without discarding the outputs of experts with relatively low contributions, as is commonly done in traditional MoE routing strategies [52].

In addition, an IWE is incorporated to provide complementary global guidance during fusion. The outputs of different RAEs may vary across regions due to the pixelwise weighting, and the IWE serves as a global reference that guides these region-adaptive contributions. This approach differs from methods such as [49], where global and local experts are explicitly separated; instead, the IWE supports each RAE to adaptively emphasize its most relevant regions.

III. Proposed Method

A. Overview of the RAMoE Network

Fig. 1 illustrates the overall architecture of the proposed network, which mainly consists of three cross-modal fusion modules (CMFMs) and SSRM. The network accepts LR-HSI $\mathbf{H} \in \mathbb{R}^{x\times y\times B}$ and HR-MSI $\mathbf{M} \in \mathbb{R}^{X\times Y\times b}$ as inputs, and the outputs are reconstructed HR-HSI $\widehat{\mathbf{H}} \in \mathbb{R}^{X\times Y\times B}$. At the beginning of the reconstruction process, $\mathbf{H}$ is upsampled via bicubic to generate an bicubic-upsampled HSI $\widetilde{\mathbf{H}} \in \mathbb{R}^{X\times Y\times B}$ that matches the spatial resolution of $\mathbf{M}$

$$\widetilde{\mathbf{H}} = \mathcal{B}(\mathbf{H}) \tag{1}$$

where $\mathcal{B}(\cdot)$ denotes the bicubic upsampling operation. Subsequently, $\widetilde{\mathbf{H}}$ and $\mathbf{M}$ are passed through convolutional layers to align their channel dimensions, generating the initialized representations of LR-HSI $\mathbf{H}_0 \in \mathbb{R}^{X\times Y\times \widetilde{B}}$ and HR-MSI $\mathbf{M}_0 \in \mathbb{R}^{X\times Y\times \widetilde{B}}$, where $\widetilde{B}$ denotes the channels of feature in the network

$$\mathbf{H}_0 = \mathcal{P}\left(\widetilde{\mathbf{H}}\right) \tag{2}$$

$$\mathbf{M}_0 = \mathcal{H}(\mathbf{M}) \tag{3}$$

where $\mathcal{P}(\cdot)$ and $\mathcal{H}(\cdot)$ represents the different convolution operation. The initialized features $\mathbf{H}_0$ and $\mathbf{M}_0$ are then subsequently processed through CMFMs to progressively integrate spatial and spectral information, enabling deep interaction and complementary feature enhancement between the hyperspectral and multispectral modalities

$$\mathbf{H}_i = \mathcal{F}_i(\mathbf{H}_{i-1}, \mathbf{M}_0) \tag{4}$$

where $i \in 1, 2, 3$. $\mathcal{F}_i(\cdot)$ represents the ith CMFM operation, and $\mathbf{H}_i \in \mathbb{R}^{X\times Y\times \widetilde{B}}$ represents the output of the ith CMFM. At the end of the network, the fusion feature of $\mathbf{H}_3$ is combined with $\widetilde{\mathbf{H}}$ and $\mathbf{M}$ for spatial and spectral reconstructions. These features are jointly processed through the SSRM to generate the final reconstructed HR-HSI $\widehat{\mathbf{H}} \in \mathbb{R}^{X\times Y\times B}$

$$\widehat{\mathbf{H}} = \mathcal{S}\left(\widetilde{\mathbf{H}}, \mathbf{M}, \mathbf{H}_3\right) \tag{5}$$

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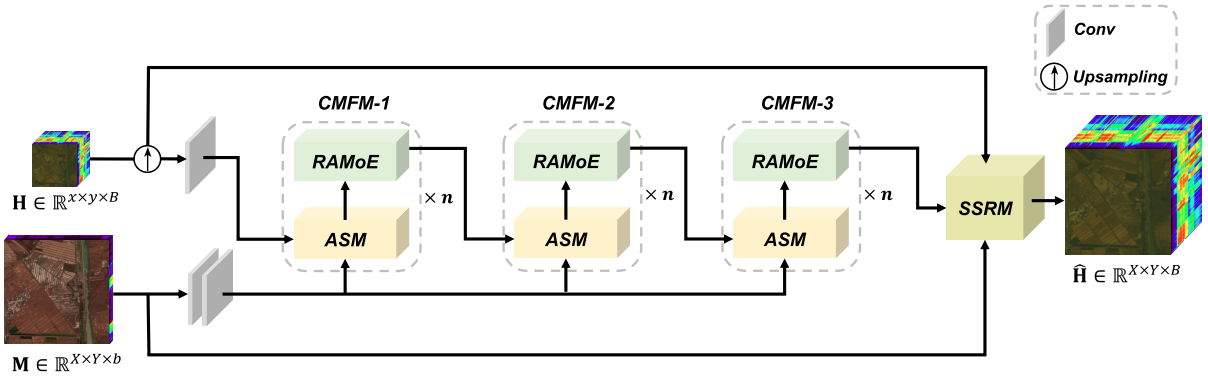


Fig. 1. Overview of the proposed RAMoE network architecture. The LR-HSI $\mathbf{H}$ and the HR-MSI $\mathbf{M}$ are jointly processed through three CMFMs. Each CMFM comprises n sequentially stacked blocks, where each block incorporates an ASM followed by the RAMoE. The fused features are finally refined by the SSRM to produce the reconstructed HR-HSI $\widehat{\mathbf{H}}$.

where $\mathcal{S}(\cdot)$ represents the SSRM operation. In summary, the proposed network performs joint spatial and spectral enhancement through a hierarchical cross-modal fusion framework and a spatial and spectral reconstruction mechanism. By effectively exploiting the complementary characteristics of hyperspectral and multispectral modalities, the proposed network exhibits superior capability in preserving fine spatial structures and maintaining high spectral fidelity in the reconstructed HSIs.

B. Cross-Modal Fusion Module

As illustrated in Fig. 1, each CMFM contains n sequential subblocks, and each subblock is composed of two essential components: the ASM and the RAMoE. The ASM is designed to share attention maps between HSI and MSI modalities, enabling effective interaction and complementary feature utilization across both sources, while the RAMoE employs a weighting mechanism to integrate multiple expert features, thereby improving the expressiveness and robustness of the fused representations.

Within this module, the initialized MSI features $\mathbf{M}_0$ remain fixed across all CMFM, serving as a stable source of high-resolution spatial information. In contrast, the HSI feature $\mathbf{H}_i$ is progressively updated after being fused with $\mathbf{M}_0$ in the first CMFM; the resulting feature is propagated into the next CMFM for further refinement. With each successive CMFM, the interaction between HSI and MSI features becomes increasingly deep and complementary. This progressive fusion strategy enables a gradual and hierarchical integration of spatial and spectral cues, effectively mitigating modality discrepancies. Moreover, by continuously refining the hyperspectral representation under the guidance of the stable MSI features, the network can better preserve spatial consistency while enhancing spatial and spectral precision in the reconstructed HSI.

C. Attention Sharing Module

In the LR-HSI and HR-MSI fusion task, the input data are acquired from different sensors. Traditional transformer-based approaches that process a single image modality often fail to exploit the complementary information contained in the other modality, thereby limiting effective interaction between HSI and MSI features and potentially degrading overall performance. To address this issue and enhance cross-modal feature fusion, we modify the standard self-attention computation by introducing an attention sharing mechanism, in which the attention operation jointly incorporates information from both HSI and MSI. The detailed structure of the proposed ASM is illustrated in Fig. 2. Given $\mathbf{H}_i$ and $\mathbf{M}_0$ as inputs, the ASM first applies separate convolutional layers to each input to extract modality-specific feature representations. These extracted features are then fed into a multihead attention mechanism. For each attention head, the query ($\mathbf{Q}_1$, $\mathbf{Q}_2$), key ($\mathbf{K}_1$, $\mathbf{K}_2$), and value ($\mathbf{V}_1$, $\mathbf{V}_2$) matrices are derived from the corresponding input features $\mathbf{H}_i$ and $\mathbf{M}_0$ through learned nonlinear convolutional projection layers, where $\mathbf{Q}_1, \mathbf{Q}_2, \mathbf{K}_1, \mathbf{K}_2, \mathbf{V}_1, \mathbf{V}_2 \in \mathbb{R}^{X\times Y\times \widetilde{B}}$

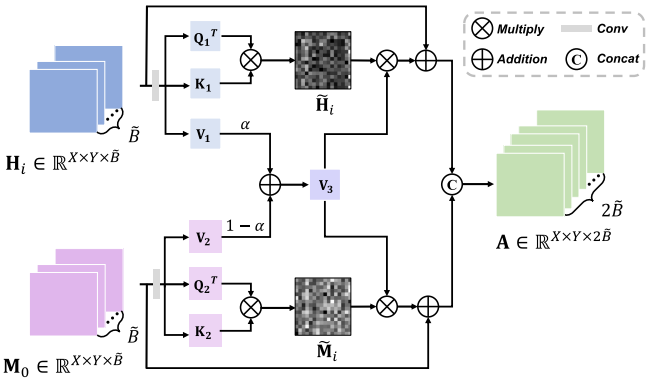


Fig. 2. Flowchart of the proposed ASM. The input $\mathbf{H}_i$ denotes the iteratively fused feature at the ith stage, while $\mathbf{M}_0$ represents the initial feature extracted from the HR-MSI. A learnable scalar parameter α is introduced to adaptively combine $\mathbf{V}_1$ and $\mathbf{V}_2$, facilitating effective cross-modal interaction.

$$\begin{aligned}\mathbf{Q}_1, \mathbf{K}_1, \mathbf{V}_1 &= \mathcal{U}(\mathbf{H}_i)\\ \mathbf{Q}_2, \mathbf{K}_2, \mathbf{V}_2 &= \mathcal{V}(\mathbf{M}_0)\end{aligned} \tag{6}$$

where $\mathcal{U}(\cdot)$ and $\mathcal{V}(\cdot)$ denote different convolution operations. Subsequently, $\mathbf{V}_1$ and $\mathbf{V}_2$, derived from different inputs, are combined through a learnable weighting mechanism to produce a new value matrix $\mathbf{V}_3 \in \mathbb{R}^{X\times Y\times \widetilde{B}}$

$$\mathbf{V}_3 = \alpha \mathbf{V}_1 + (1-\alpha)\mathbf{V}_2 \tag{7}$$

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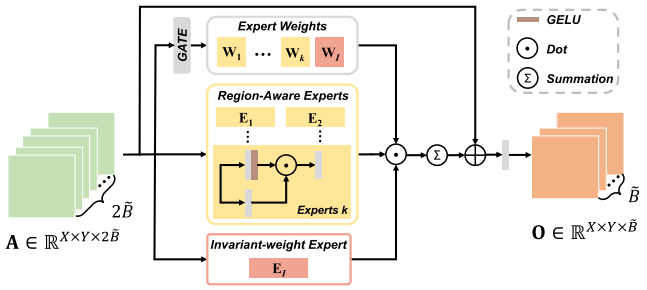


Fig. 3. Flowchart of the proposed region-aware MoE (RAMoE). The RAEs assign pixelwise adaptive weights, while the IWE applies a uniform weight across the entire image, thereby providing globally consistent guidance.

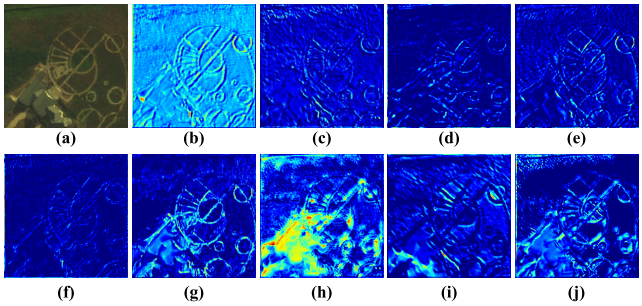


Fig. 4. Visualization of the fusion result and different expert outputs. (a) Pseudocolor representation of the fusion result. (b) Output of the IWE. (c)–(j) Outputs of the RAEs. Note that the IWE (b) highlights high-frequency structures such as object boundaries, while the RAEs (c)–(j) exhibit activation in complementary spatial regions, reflecting their region-aware specialization.

where α is a learnable scalar parameter. The parameter α is optimized jointly with the rest of the network during training and is used to adaptively regulate the relative contributions of $\mathbf{V}_1$ and $\mathbf{V}_2$ in forming $\mathbf{V}_3$. Then, the attention maps of $\mathbf{H}_i$ and $\mathbf{M}_0$ are computed by

$$\widetilde{\mathbf{H}}_i = \mathbf{V}_3 \mathcal{T}\left(\mathbf{Q}_1^T \mathbf{K}_1 \cdot \beta\right)$$
$$\widetilde{\mathbf{M}}_0 = \mathbf{V}_3 \mathcal{T}\left(\mathbf{Q}_2^T \mathbf{K}_2 \cdot \gamma\right) \tag{8}$$

where $\mathcal{T}(\cdot)$ denotes the softmax function, and β and γ are the learnable scalar parameters. Finally, $\widetilde{\mathbf{H}}_i \in \mathbb{R}^{X\times Y\times \widetilde{B}}$ and $\widetilde{\mathbf{M}}_0 \in \mathbb{R}^{X\times Y\times \widetilde{B}}$ are applied to obtain the unified attention features $\mathbf{A} \in \mathbb{R}^{X\times Y\times 2\widetilde{B}}$

$$\mathbf{A} = \mathcal{C}\left[\widetilde{\mathbf{H}}_i + \mathbf{H}_i \,\|\, \widetilde{\mathbf{M}}_0 + \mathbf{M}_0\right] \tag{9}$$

where $\mathcal{C}[\cdot \,\|\, \cdot]$ represents the channelwise concatenation operator. Through this mechanism, the attention extracted from HSI and MSI is effectively shared across modalities, forming a consistent attention representation. Importantly, the full attention features are retained rather than compressed or discarded, allowing the following RAMoE module to further model and refine the cross-modal interactions.

D. Region-Aware MoE

As shown in Fig. 3, the proposed RAMoE architecture comprises a gating network, a set of region-aware experts (RAEs), and the IWE. During training, different experts produce heterogeneous feature representations, and the quality of these features may vary across different spatial regions. Therefore, it is necessary to emphasize the regions where the feature quality is comparatively higher. Guided by the gating mechanism, the experts assign adaptive weights to the expert outputs across spatial locations such that regions with higher-quality features receive greater emphasis. As a result, each expert becomes more specialized in the regions it represents well, leading to more reliable and discriminative feature fusion across the image.

In RAMoE, the input feature $\mathbf{A}$ produced by the ASM is processed in parallel by the gating network, the RAEs, and the IWE. Specifically, the gating network generates a set of weight maps $\{\mathbf{W}_1, \mathbf{W}_2, \ldots, \mathbf{W}_k, \mathbf{W}_I\}$, while the RAEs produce a collection of feature representations $\{\mathbf{E}_1, \mathbf{E}_2, \ldots, \mathbf{E}_k\}$, and the IWE outputs the feature representation $\mathbf{E}_I$. All expert features satisfy $\mathbf{E}_i, \mathbf{E}_I \in \mathbb{R}^{X\times Y\times 2\widetilde{B}}$, and the corresponding weight maps satisfy $\mathbf{W}_i, \mathbf{W}_I \in \mathbb{R}^{X\times Y\times 2\widetilde{B}}$. The weights associated with the RAEs, $\{\mathbf{W}_1, \mathbf{W}_2, \ldots, \mathbf{W}_k\}$, are adaptively generated from the input features and vary across spatial locations, enabling pixelwise modulation of expert contributions. In contrast, $\mathbf{W}_I$ corresponds to the IWE and is defined as a spatially fixed weight map that assigns identical weights to all pixel locations, thereby providing a stable global contribution during feature aggregation. Specifically, since $\mathbf{W}_I$ is spatially invariant, the IWE produces a uniform feature response across the entire image, maintaining a consistent baseline representation throughout all spatial locations. This globally consistent response supplies stable contextual information and helps prevent the RAEs from overspecializing in highly localized patterns. By acting as a global reference during aggregation, the IWE complements the spatially adaptive modeling of the RAEs and promotes a more balanced fusion process that preserves global structural information while allowing the RAEs to enhance region-specific details. All experts employ an identical network architecture and produce features in the same representation space, differing only in their associated weighting strategies. This design facilitates feature compatibility across experts and combines spatially adaptive weighting with a globally consistent expert contribution, thereby preserving global structural coherence while enabling flexible enhancement of local details and avoiding hard routing strategies that discard expert outputs. Based on the learned weights, the expert features are adaptively aggregated to produce the final region-fused representation, denoted as $\widetilde{\mathbf{E}} \in \mathbb{R}^{X\times Y\times 2\widetilde{B}}$

$$\widetilde{\mathbf{E}} = \sum_{j=1}^{k} \left(\mathbf{W}_j \times \mathbf{E}_j\right) + \mathbf{W}_I \times \mathbf{E}_I \tag{10}$$

where k denotes the number of the RAEs.

To better illustrate the distinction between the IWE and the RAEs with spatially adaptive weights in RAMoE, Fig. 4 visualizes the weighted expert outputs, i.e., $\mathbf{W}_j \times \mathbf{E}_j$, for different experts. The IWE exhibits strong responses across the entire image, reflecting its global attention over all spatial regions. In contrast, the other experts with spatially adaptive weights attend to different and complementary regions, indicating distinct regional preferences. Furthermore, the visualization of the IWE suggests that, without spatially adaptive weight, this expert tends to emphasize high-frequency components such as object boundaries, thereby capturing rich and

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globally consistent texture information. By comparison, the other experts with spatially adaptive weights focus on different spatial regions, enabling region-dependent feature enhancement. This complementary behavior demonstrates the effectiveness of integrating invariant and spatially adaptive experts within the proposed RAMoE.

The final output $\mathbf{O} \in \mathbb{R}^{X\times Y\times \widetilde{B}}$ is further processed with a residual connection, and a subsequent convolution layer is employed to align the channel dimension with that of $\mathbf{H}_i$, ensuring structural consistency

$$\mathbf{O} = \mathcal{I}\left(\widetilde{\mathbf{E}} + \mathbf{A}\right) \tag{11}$$

where $\mathcal{I}(\cdot)$ denotes the convolution operation. The gating network assigns expert weights in a pixelwise manner, allowing the fusion process to be carried out at the pixel level rather than globally. This pixel-level adaptive fusion effectively captures fine-grained spatial variations in modality contributions. Moreover, the multiexpert strategy endows the module with more comprehensive feature representation capability and stronger robustness when handling complex fusion tasks. By integrating the complementary information learned by different experts, RAMoE not only improves the quality of fusion results but also enhances the model's generalization ability across various scenarios and data conditions.

Algorithm 1 Pseudocode of the Proposed Method

Input: 1) LR-HSI $\mathbf{H} \in \mathbb{R}^{x\times y\times B}$, 2) HR-MSI $\mathbf{H} \in \mathbb{R}^{X\times Y\times b}$
Output: Reconstructed HR-HSI $\widehat{\mathbf{H}} \in \mathbb{R}^{X\times Y\times B}$

1: **Step 1: Get initialized representation of LR-HSI and HR-MSI**
2: Upsample LR-HSI using bicubic interpolation to obtain $\widetilde{\mathbf{H}} \in \mathbb{R}^{X\times Y\times B}$ (via Eq.(1)).
3: Generating the initialized representations $\mathbf{H}_0 \in \mathbb{R}^{X\times Y\times \widetilde{B}}$ and $\mathbf{M}_0 \in \mathbb{R}^{X\times Y\times \widetilde{B}}$ from $\widetilde{\mathbf{H}}$ and $\mathbf{M}$ (via Eq.(2) and (3)).
4: **Step 2: Cross-modal fusion**
5: Get attention maps $\widetilde{\mathbf{H}}_i \in \mathbb{R}^{X\times Y\times \widetilde{B}}$ and $\widetilde{\mathbf{M}}_0 \in \mathbb{R}^{X\times Y\times \widetilde{B}}$ by the mixed value matrix $\mathbf{V}_3 \in \mathbb{R}^{X\times Y\times \widetilde{B}}$ (via Eq.(6), (7) and (8)).
6: Get unified attention $\mathbf{A} \in \mathbb{R}^{X\times Y\times 2\widetilde{B}}$ by concat $\widetilde{\mathbf{H}}_i$ and $\widetilde{\mathbf{M}}_0$ with residual connections (via Eq.(9)).
7: Compute fused expert outputs $\widetilde{\mathbf{E}} \in \mathbb{R}^{X\times Y\times 2\widetilde{B}}$ from RAE and IWE using the gating network (via Eq.(10)).
8: Align the channel dimension to obtain an intermediate result, which is used as the input of the next CMFM (via Eq. (11)).
9: **Step 3: Spatial and Spectral Reconstruction**
10: Reconstruct the $\widehat{\mathbf{H}}$ with original input $\mathbf{H}$ and $\mathbf{M}$ (via Eq.(12)).

E. Spatial and Spectral Reconstruction Module

The LR-HSI contains rich spectral information but lacks spatial details, while the HR-MSI provides fine spatial structures but suffers from limited spectral diversity. To reconstruct the HR-HSI, the SSRM aims to restore the missing high-frequency spectral and spatial components by estimating residual information that compensates for the spatial and spectral degradation in $\mathbf{H}_3$. Considering that the HR-MSI captures complementary spatial information correlated with the residual details absent in the LR-HSI, it is natural to incorporate the HR-MSI into the residual estimation process. By jointly utilizing the features from both $\widetilde{\mathbf{H}}$ and $\mathbf{M}$, the SSRM can recover the fine spatial structures while preserving the spectral consistency of the reconstructed image. Fig. 5 shows the architecture of the proposed SSRM. Specifically, we concatenate $\widetilde{\mathbf{H}}$ and $\mathbf{M}$ along the channel dimension and apply a shallow feature extraction module, which consists of three convolutional layers with 3×3 kernels and stride 1. The final fusion result $\widehat{\mathbf{H}}$ is obtained by adding the band-aligned output of the third CMFM to the extracted shallow feature, as formulated by

$$\widehat{\mathbf{H}} = \mathcal{A}\left(\mathbf{H}_3\right) + \mathcal{M}\left(\mathcal{C}\left[\widetilde{\mathbf{H}} \,\|\, \mathbf{M}\right]\right) \tag{12}$$

where $\mathcal{C}[\cdot \,\|\, \cdot]$ denotes the channelwise concatenation operator, $\mathcal{M}(\cdot)$ represents the shallow feature extraction operation, and $\mathcal{A}(\cdot)$ denotes the convolution layer that aligns the channel dimension of $\mathbf{H}_3$ to the B. This design allows the SSRM to effectively leverage spatial priors from the HR-MSI to enhance structural details while maintaining the spectral fidelity inherited from the LR-HSI. The pseudocode of the proposed method is provided in Algorithm 1.

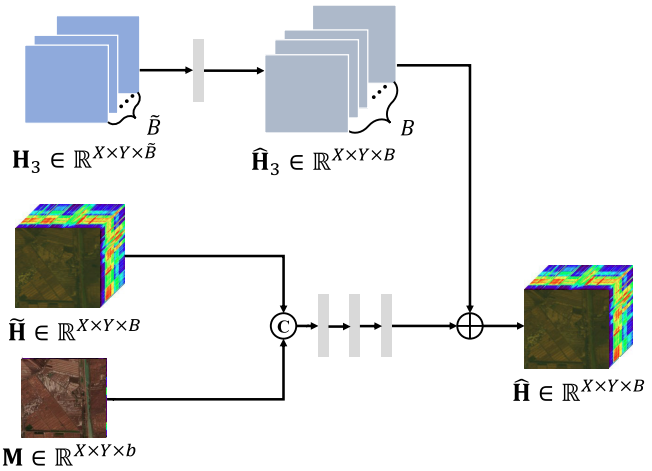


Fig. 5. Flowchart of the proposed SSRM. The upsampled LR-HSI and the HR-MSI are concatenated to extract residual information, which is then added to the band-aligned output of the third CMFM to produce the final fused result, enhancing spatial details while preserving spectral fidelity.

F. Loss Function

To optimize the network parameters, we employ the L_1 loss function, which measures the absolute difference between the reconstructed HR-HSI $\widehat{\mathbf{H}}$ and the ground truth $\mathbf{H}^*$. The loss is defined as

$$\mathcal{L}_1 = \frac{1}{N}\sum_{i=1}^{N}\left|\widehat{\mathbf{H}}_i - \mathbf{H}_i^*\right| \tag{13}$$

where N denotes the total number of training samples and $|\cdot|$ represents the L_1 norm.

Compared with the $\mathcal{L}_2$ loss, the $\mathcal{L}_1$ loss is less sensitive to outliers and better preserves the structural and edge details of the reconstructed image. This property is particularly beneficial for HSI fusion, as it helps maintain the spatial sharpness

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and spectral consistency while preventing oversmoothing in the recovered high-frequency components.

IV. Experimental Results

In this section, we first introduce the four datasets used. Then, we describe the experimental settings, including the data simulation, implementation details, evaluation metrics, and comparison methods. Finally, ablation experiments are performed to validate the effectiveness of our proposed modules, and experimental results on the four datasets used are shown.

A. Datasets

The comparison experiments are conducted on four datasets, including Washington DC Mall (WDCM) [53], Chikusei [54], Xiongan [45], and Yellow River Estuary (YRE) [55].

1) Washington DC Mall: The WDCM dataset was obtained by the sensor of the hyperspectral digital imagery collection experiment (HYDICE) over the National Mall in Washington, DC, in 1995. In WDCM, there are 210 bands in total that cover the wavelengths from 0.4 to 2.5 μm. After removing the bands of water vapor absorption, the remaining 191 bands are used in this article, and the image of each band measures 1280×307 pixels with a spatial resolution of 2.5 m.

2) Chikusei: The Chikusei dataset was obtained from the Hyperspec-VNIR-C imaging sensor in Chikusei, Japan. The spatial resolution of HSI is 2.5 m, with a size of 2517×2335, and includes 128 spectral bands spanning from 363 to 1018 nm. When processing the HSI data, we excluded the bands with low signal-to-noise ratio, thereby reducing the number of bands to 110.

3) Xiongan: The Xiongan dataset was collected by the full-spectrum multimodal imaging spectrometer of the Gaofen Special Aviation System developed by the Shanghai Institute of Technical Physics, Chinese Academy of Sciences. The original size of the Xiongan dataset is $3750 \times 1580 \times 250$, with each pixel representing 0.5 m, and the spectral bands range from 400 to 1000 nm. We reduced the number of bands to 93 for computational efficiency.

4) Yellow River Estuary: The YRE dataset consists of HSI and MSI from different satellites. Specifically, the HSI is acquired by the advanced hyperspectral sensor onboard the Gaofen-5 satellite, while the MSI is obtained from the multispectral sensor onboard the Sentinel-2A satellite. The images cover the eastern region of the Yellow River Delta in China. The HSI has a spatial resolution of 30 m, with a spatial size of 1400×1400 pixels, and 110 spectral bands range from 400 to 1200 nm. In contrast, the MSI has a spatial resolution of 10 m, with a spatial size of 4200×4200 pixels and four spectral bands covering the visible wavelength range from 430 to 680 nm.

B. Experimental Settings

1) Data Simulation: For the three simulated datasets (WDCM, Chikusei, and Xiongan), the original HSI is used as the reference image. To obtain the LR-HSI, a Gaussian filter with a kernel size of 5×5 and a standard deviation of 2 is applied to the reference image, followed by downsampling by a factor of 4 and 8 to obtain different spatial resolution LR-HSI. To obtain the HR-MSI, in the WDCM dataset, the HR-MSI with ten bands was generated using the spectral response matrix of the Sentinel-2 A instrument [56]. For the Chikusei dataset, the HR-MSI with five bands was generated using the spectral response matrix from YOKOYA [57]. For the Xiongan dataset, the HR-MSI with four bands was generated following Wang et al. [45]. The training images are divided into patches of 64×64 pixels. For each dataset, the data were divided along the height dimension into two nonoverlapping subsets, with 70% used for training and 30% for testing.

2) Implementation Details: In our implementation, the feature channel $\widetilde{B}$ is set as 96, and the number n in CMFM is set as 3. In RAMoE, the number of RAEs is set to $k = 8$, and one IWE is employed with its weight fixed to 1. All experiments covered in this article are based on the PyTorch framework on Python 3.9. We used the Adam optimizer with an initial learning rate of $2e^{-4}$. The model was trained for a total of 500 epochs, starting from the 150th epoch, and the learning rate was halved every 50 epochs to further improve the performance. All experiments are performed on a computer with an Intel Xeon Gold 6430 CPU and an Nvidia A100 GPU, 80-GB RAM.

3) Evaluation Metrics: Three commonly used evaluation metrics are employed to assess the quality of the fused HSI produced by different methods, which include the peak signal-to-noise ratio (PSNR) [58], the spectral angle mapper (SAM) [59], and the relative dimensionless global error in synthesis (ERGAS) [58]. Higher PSNR indicates better image quality, while lower ERGAS and SAM values suggest improved performance. These metrics offer a comprehensive evaluation of the fused HSI in terms of both spectral fidelity and spatial quality. In real-data fusion experiments, the nonreference metrics, D_λ, D_s, and QNR [60], are used for evaluation. D_λ and D_s measure spectral and spatial distortions, respectively, with lower values indicating better performance. QNR jointly reflects the spectral and spatial quality computed from D_λ and D_s, with a maximum value of 1.

4) Comparison Methods: We selected eight state-of-the-art LR-HSI and HR-MSI fusion methods for comparison, including CNN-based methods and transformer-based methods. For CNN-based methods, we included several methods, such as SSRNet [38], ResTFNet [39], MOG-DCN [40], DMSN [61], and FusGAT [62]. For transformer-based approaches, we selected state-of-the-art methods including Fusformer [42], DSPNet [43], 3DT-Net [44], DCTransformer [30], PSTUN [45], CYformer [63], and DTUML [64]. To ensure a fair comparison, all deep learning-based methods, including the proposed method, have been retrained on the same training dataset. The results were obtained by running the official source codes downloaded from the authors' website.

C. Ablation Studies

In this section, we conduct comprehensive ablation studies to evaluate the contribution of each key component in our

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TABLE I

ABLATION STUDY ON THE NUMBER OF RAE IN RAMOE. THE OPTIMAL VALUES ARE SHOWN IN BOLD

Number of RAE	PSNR↑	SAM↓	ERGAS↓
4	50.89	0.97	0.3322
6	51.45	0.92	0.3112
8	**51.92**	**0.89**	**0.2956**
10	51.18	0.96	0.3212
12	50.83	1.01	0.3358

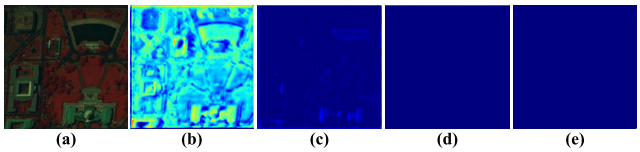


Fig. 6. Visualization of the fusion result and IWE outputs. (a) Pseudocolor representation of the fusion result. (b)–(e) Outputs of the IWE.

TABLE II

ABLATION STUDY ON THE NUMBER OF IWES IN RAMOE. THE OPTIMAL VALUES ARE SHOWN IN BOLD

Number of IWE	PSNR↑	SAM↓	ERGAS↓
0	51.92	0.89	0.2956
1	**52.46**	**0.84**	**0.2783**
2	52.33	0.85	0.2822
3	52.16	0.87	0.2878

network. All ablation experiments are conducted on the WDCM dataset with a scaling factor of 8.

We first investigate the effect of varying the number of (RAEs). As shown in Table I, the performance consistently improves as the number of RAEs increases from 4 to 8, where the model achieves the highest PSNR, lowest SAM, and lowest ERGAS. This indicates that a moderate number of experts provides sufficient feature diversity, enabling the model to better capture spatial and spectral variations and fuse them. However, when the number of experts continues to grow beyond eight, the performance begins to degrade. This suggests that excessive experts introduce redundant feature representations, which, in turn, increases the complexity of the fusion process and weakens the effectiveness of the gating mechanism. Therefore, in the following experiments, the number of RAEs in RAMoE is fixed to 8.

Furthermore, Table II demonstrates the effectiveness of introducing the IWE into RAMoE. Compared with the baseline without IWE, incorporating a single IWE consistently improves all evaluation metrics, indicating that the IWE effectively enhances the overall fusion performance. However, when more IWEs are introduced, the performance shows a slight decline, suggesting that increasing the number of IWEs does not lead to further performance gains. To further illustrate this phenomenon, Fig. 6 visualizes the outputs of IWE when four IWEs are employed. Since all IWEs share fixed and identical weights, their outputs exhibit highly similar characteristics and do not form complementary behaviors, unlike the RAEs that attend to different spatial regions. As a result, introducing multiple IWEs mainly increases redundancy rather than enhancing representational diversity, which may

TABLE III

ABLATION STUDY ON THE IMPACT OF DIFFERENT INPUT FEATURES IN SSRM. THE OPTIMAL VALUES ARE SHOWN IN BOLD

$\mathbf{H}_3$	$\widetilde{\mathbf{H}}$	$\mathbf{M}$	PSNR↑	SAM↓	ERGAS↓
✓			51.32	0.95	0.3167
✓	✓		51.91	0.87	0.2949
✓		✓	51.82	0.92	0.2992
✓	✓	✓	**52.46**	**0.84**	**0.2783**

slightly weaken the balance between global consistency and region-adaptive modeling. These observations indicate that a single IWE is sufficient to provide effective global guidance. Therefore, in the subsequent experiments, we adopt one IWE in RAMoE.

To examine the influence of inputting different residual information in the SSRM, we evaluate four configurations that differ in the input features provided to the SSRM, as reported in Table III. When no residual information is used, the network relies solely on the fused representation from $\mathbf{H}_3$ and achieves poor evaluation metrics. The inclusion of the upsampled LRHSI as a residual input effectively refines the reconstruction, yielding 51.91 dB in terms of PSNR and a reduced SAM, which demonstrates its benefit in preserving spectral consistency. However, although $\widetilde{\mathbf{H}}$ provides spectral priors, its spatial structures deviate considerably from the ground truth, which limits its effectiveness. Although using $\mathbf{M}$ alone as residual information leads to a considerable increase in PSNR, the improvement in SAM is marginal, revealing that $\mathbf{M}$ contributes mainly to spatial details but lacks effective spectral information, thus limiting the overall reconstruction accuracy. In contrast, when both $\widetilde{\mathbf{H}}$ and $\mathbf{M}$ are used jointly, SSRM effectively integrates the spectral guidance from $\widetilde{\mathbf{H}}$ with the spatial information from $\mathbf{M}$, resulting in more accurate hyperspectral reconstruction. This complementary residual representation enables more accurate spectral correction and spatial detail reconstruction, yielding the best overall performance with 52.46 dB in terms of PSNR, and the lowest SAM and ERGAS values. These results demonstrate that the spectral and spatial residual information provided by SSRM is crucial for maintaining spectral fidelity while reconstructing high-spatial-resolution HSIs.

In order to evaluate the contributions of the ASM, RAMoE, and SSRM modules to the performance of the proposed network, we construct a series of network variants by progressively removing these three components. Specifically, we examine configurations without all module, with only one module retained, with pairwise combinations of the modules, and finally with all three components integrated. It is worth noting that without ASM, the attention is computed independently within each modality, resulting in the absence of cross modality feature interaction. When RAMoE is removed, the RAMoE mechanism collapses to a single expert, thereby losing region-aware adaptive modeling ability. Moreover, without SSRM, no additional residual spatial and spectral information is added to the network.

Table IV summarizes the quantitative results of the ablation experiments under different configurations. The baseline network without any of the proposed components exhibits the

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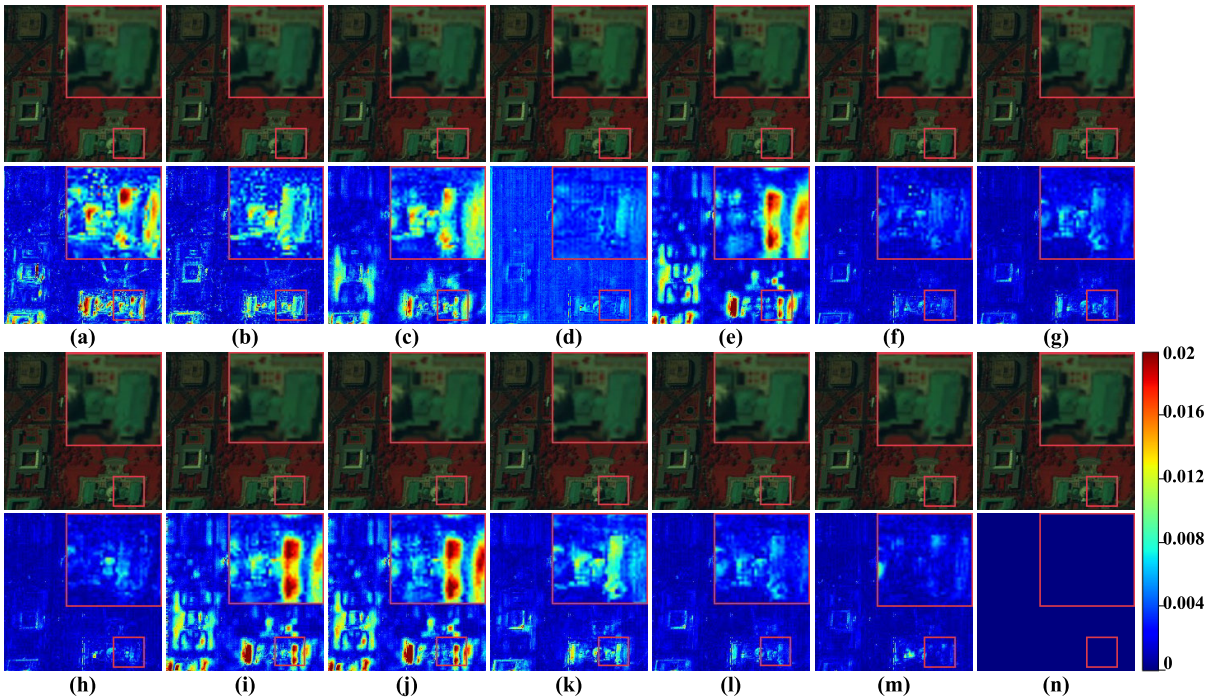


Fig. 7. Qualitative comparison of different methods on the WDCM dataset. The first row shows the results using the pseudocolor representation (R-65, G-30, and B-8), and the second row is the corresponding error images. (a) SSRNet. (b) ResTFNet. (c) MoG-DCN. (d) Fusformer. (e) DSPNet. (f) 3DT-Net. (g) DCTransformer. (h) PSTUN. (i) CYformer. (j) DMSN. (k) FusGAT. (l) DTUML. (m) Ours. (n) Reference.

TABLE IV

ABLATION STUDY ON THE EFFECTS OF ASM, GARMOE, AND SRM COMPONENTS. THE OPTIMAL VALUES ARE SHOWN IN BOLD

ASM	GARMoE	SRM	PSNR↑	SAM↓	ERGAS↓
			49.05	1.21	0.4104
✓			50.14	1.09	0.3626
	✓		50.33	1.06	0.3521
		✓	50.02	1.11	0.3672
✓	✓		51.32	0.95	0.3167
✓		✓	51.55	0.93	0.3074
	✓	✓	51.96	0.88	0.2923
✓	✓	✓	**52.46**	**0.84**	**0.2783**

lowest reconstruction quality, achieving only 49.05 dB in terms of PSNR, along with the highest SAM and ERGAS values. When the ASM is introduced individually, the performance improves to 50.14 dB in terms of PSNR. This improvement suggests that the attention maps extracted from HSI and MSI exhibit strong complementarity. The shared attention mechanism enables more coherent intermodal correlation modeling and facilitates more effective cross-modal feature interaction. Incorporating the RAMoE yields a further performance gain, reaching 50.33 dB in terms of PSNR, suggesting that the expert selection mechanism is beneficial for adaptively capturing region-varying structural characteristics. Likewise, the introduction of the SSRM alone leads to 50.02 dB in terms of PSNR, confirming that explicitly leveraging the original HSI and MSI provides additional spectral guidance that reduces reconstruction distortion. More substantial improvements are obtained when two components are used jointly. For example, the combination of ASM and RAMoE increases to 51.32 dB in terms of PSNR, while the integration of ASM and SSRM results in 51.55 dB in terms of PSNR, and RAMoE together with SSRM achieves 51.96 dB in terms of PSNR. The complete model incorporating ASM, RAMoE, and SSRM achieves the best overall performance, with the highest PSNR and the lowest SAM and ERGAS. These results confirm that the concurrent enforcement of attention sharing, expert-weighted feature integration, and spectral reconstruction provides the most comprehensive enhancement in both spatial detail preservation and spectral fidelity.

D. Comparison Experiments on Simulated Data

In this section, we present comprehensive qualitative and quantitative comparisons between the proposed method and several state-of-the-art methods to validate the effectiveness of the proposed method in LR-HSI and HR-MSI fusion. The fusion results of the PSNR, SAM, and ERGAS on the three simulated datasets are shown in Tables V–VII. The visual results of the fused images with the corresponding error images are illustrated in Figs. 7–9.

As shown in Table V, the proposed method consistently achieves superior performance across both $\times 4$ and $\times 8$ super-resolution settings. Compared with CNN- and transformer-based methods, the proposed approach achieves higher PSNR values while simultaneously obtaining lower SAM and ERGAS scores, indicating improved spatial detail recovery together with enhanced spectral fidelity. Notably, this performance advantage remains stable as the upsampling scale increases from $\times 4$ to $\times 8$, suggesting that the proposed cross-modal fusion and reconstruction strategy is robust to larger resolution gaps. By contrast, several comparison methods show more evident performance degradation at higher scaling factors, particularly with respect to spectral

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TABLE V

QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE WDCM DATASET UNDER ×4 AND ×8 SUPER-RESOLUTION RATIOS. THE BEST RESULTS ARE SHOWN IN BOLD

Method \ Scale	PSNR↑		SAM↓		ERGAS↓	
	×4	×8	×4	×8	×4	×8
SSRNet	46.48	46.45	1.65	1.64	1.0922	0.5482
ResTFNet	46.19	45.78	1.72	1.80	1.1374	0.5956
MoG-DCN	50.57	48.45	1.04	1.29	0.6866	0.4378
Fusformer	50.15	48.22	1.09	1.34	0.9989	0.4482
DSPNet	47.22	45.64	1.52	1.83	1.0097	0.6067
3DT-Net	52.70	51.37	0.81	0.95	0.5379	0.3303
DCTransformer	52.88	51.17	0.80	0.97	0.5289	0.3212
PSTUN	51.20	51.05	0.96	0.98	0.6393	0.3253
CYformer	50.50	49.84	1.04	1.12	0.6903	0.3730
DMSN	50.71	50.33	1.00	1.06	0.6731	0.3526
FusGAT	51.22	49.07	0.95	1.21	0.6364	0.4082
DTUML	52.64	50.93	0.82	0.99	0.5430	0.3301
Ours	**53.93**	**52.46**	**0.72**	**0.84**	**0.4752**	**0.2783**

TABLE VI

QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE CHIKUSEI DATASET UNDER ×4 AND ×8 SUPER-RESOLUTION RATIOS. THE BEST RESULTS ARE SHOWN IN BOLD

Method \ Scale	PSNR↑		SAM↓		ERGAS↓	
	×4	×8	×4	×8	×4	×8
SSRNet	46.21	44.88	0.99	1.16	0.5874	0.3458
ResTFNet	44.22	43.27	1.26	1.41	0.7482	0.4196
MoG-DCN	46.63	44.72	0.94	1.18	0.5612	0.3518
Fusformer	46.75	45.39	0.92	1.03	0.5539	0.3025
DSPNet	45.60	43.80	1.07	1.33	0.6360	0.3960
3DT-Net	47.05	45.66	0.90	1.05	0.5343	0.3155
DCTransformer	47.54	46.03	0.82	1.02	0.5042	0.3042
PSTUN	47.40	45.83	0.87	1.05	0.5143	0.3108
CYformer	45.74	44.26	1.05	1.26	0.6231	0.3735
DMSN	46.09	44.67	1.01	1.20	0.5987	0.3559
FusGAT	46.95	45.29	0.89	1.11	0.5402	0.3316
DTUML	47.23	45.72	0.87	1.05	0.5228	0.3152
Ours	**48.10**	**47.21**	**0.79**	**0.88**	**0.4713**	**0.2629**

accuracy and structural consistency. As shown in Fig. 7, the pseudocolor results on the WDCM dataset further illustrate the advantages of the proposed method. The reconstructed images exhibit clearer and more coherent structural details, while the corresponding error maps display smaller discrepancies relative to the reference images. Moreover, in the zoomed-in regions, the proposed method consistently produces sharper structures than the comparison methods. In particular, building regions in the enlarged areas show reduced reconstruction errors in the error maps, indicating improved structural fidelity.

The quantitative results on the Chikusei dataset are presented in Table VI. It can be observed that the proposed method outperforms the other approaches across both the ×4 and ×8 upsampling scales on all evaluation metrics, demonstrating improved spectral consistency and enhanced spatial detail recovery. As the upsampling scale increases, the proposed method exhibits a smaller performance degradation compared with the competing methods, indicating more stable reconstruction quality under larger resolution gaps.

The visual results on the Chikusei dataset are shown in Fig. 8. The Chikusei dataset mainly consists of farmland scenes dominated by large homogeneous green regions, which

TABLE VII

QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE XIONGAN DATASET UNDER ×4 AND ×8 SUPER-RESOLUTION RATIOS. THE BEST RESULTS ARE SHOWN IN BOLD

Method \ Scale	PSNR↑		SAM↓		ERGAS↓	
	×4	×8	×4	×8	×4	×8
SSRNet	48.24	46.84	0.61	0.72	0.2863	0.1686
ResTFNet	47.90	46.93	0.64	0.72	0.2994	0.1676
MoG-DCN	49.60	46.90	0.52	0.69	0.2461	0.1676
Fusformer	50.78	49.73	0.47	0.53	0.2172	0.1229
DSPNet	48.72	46.81	0.58	0.73	0.2725	0.1704
3DT-Net	50.53	48.83	0.47	0.58	0.2216	0.1354
DCTransformer	51.87	50.14	0.41	0.50	0.1915	0.1170
PSTUN	50.98	49.57	0.45	0.53	0.2113	0.1244
CYformer	49.29	46.77	0.54	0.70	0.2550	0.1706
DMSN	49.72	47.38	0.52	0.67	0.2428	0.1590
FusGAT	50.57	47.40	0.47	0.66	0.2207	0.1574
DTUML	51.49	49.87	0.42	0.52	0.1995	0.1204
Ours	**52.60**	**50.36**	**0.38**	**0.49**	**0.1758**	**0.1141**

pose challenges for recovering fine structures and subtle spectral variations. The proposed method consistently produces more accurate reconstruction results than the comparison methods. As illustrated by the error maps, our method yields noticeably smaller reconstruction errors, indicating a closer approximation to the reference image. In the zoomed-in regions, the proposed method preserves structural details more clearly and restores spatial textures with higher fidelity, further demonstrating its robustness and effectiveness in recovering accurate hyperspectral information.

E. Comparison Experiments on Real Data

The quantitative results on the Xiongan dataset, summarized in Table VII, show that our method consistently achieves the best performance among all compared approaches. The improvements in SAM and PSNR suggest that the proposed method is more effective in preserving spectral consistency while enhancing spatial reconstruction quality, indicating stable performance across different upsampling scales. The qualitative results in Fig. 9 further corroborate these quantitative findings. The selected scene contains rich and complex texture patterns, which make accurate reconstruction particularly challenging. From both the reconstructed images and the corresponding error maps, it can be observed that the proposed method yields smaller reconstruction errors, especially in texture-rich regions such as building areas, reflecting improved recovery of fine structural details. These results indicate that the proposed approach achieves a favorable balance between spatial detail restoration and spectral fidelity.

To further validate the performance of the proposed method, we perform experiments on a real dataset, i.e., the YRE dataset. The qualitative results are shown in Fig. 10. As can be observed, our method achieves better reconstruction quality, exhibiting clearer spatial structures and more faithful texture restoration. Compared with other approaches, our results are visually closer to the HR-MSI reference, particularly in preserving high-frequency details. In the zoomed-in regions, the fused results display farmland boundaries with greater clarity, and the recovered spatial structures reveal details that are hardly discernible in the LR-HSI. These observations

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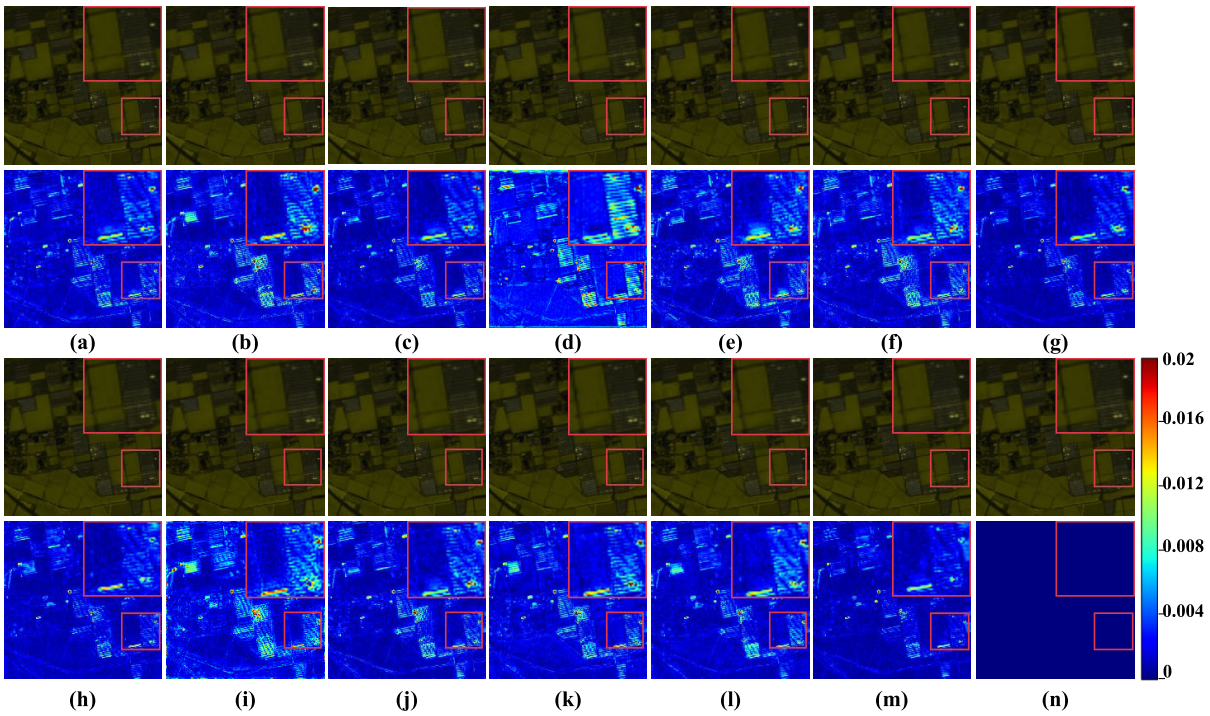


Fig. 8. Qualitative comparison of different methods on the Chikusei dataset. The first row shows the results using the pseudocolor representation (R-81, G-76, and B-2), and the second row is the corresponding error images. (a) SSRNet. (b) ResTFNet. (c) MoG-DCN. (d) Fusformer. (e) DSPNet. (f) 3DT-Net. (g) DCTransformer. (h) PSTUN. (i) CYformer. (j) DMSN. (k) FusGAT. (l) DTUML. (m) Ours. (n) Reference.

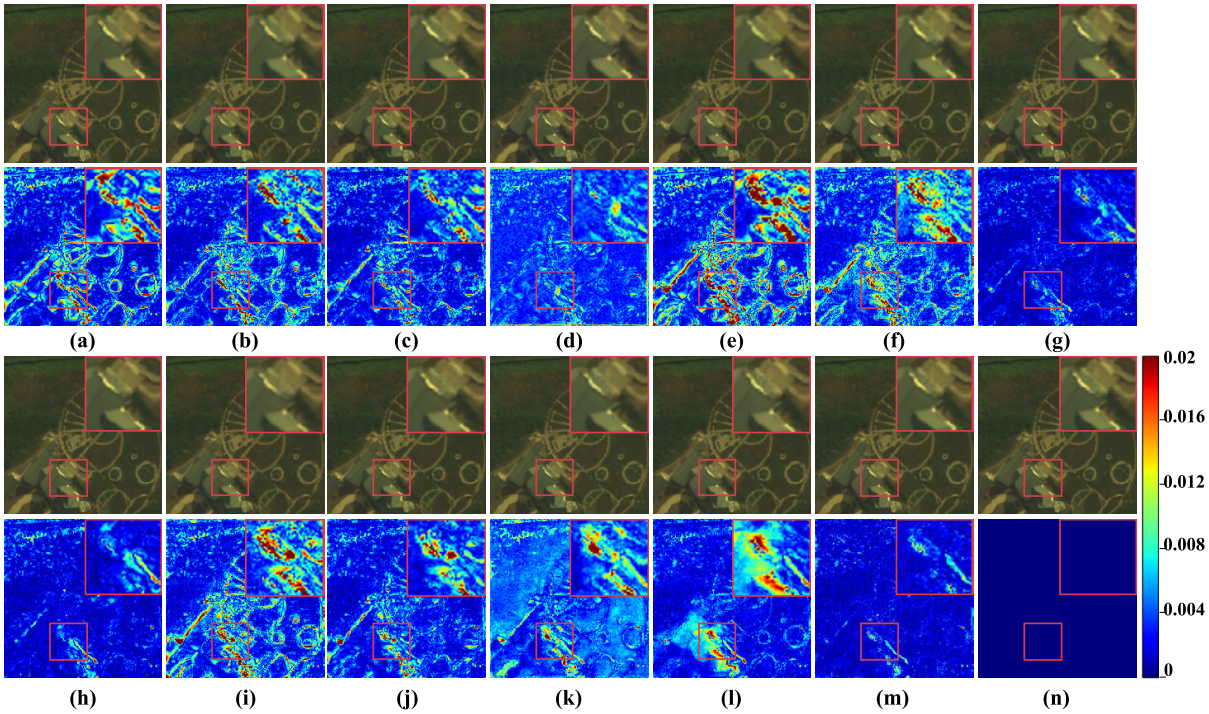


Fig. 9. Qualitative comparison of different methods on the Xiongan dataset. The first row shows the results using the pseudocolor representation (R-38, G-23, and B-8), and the second row is the corresponding error images. (a) SSRNet. (b) ResTFNet. (c) MoG-DCN. (d) Fusformer. (e) DSPNet. (f) 3DT-Net. (g) DCTransformer. (h) PSTUN. (i) CYformer. (j) DMSN. (k) FusGAT. (l) DTUML. (m) Ours. (n) Reference.

demonstrate that the proposed method is more effective in restoring real-scene geometric and textural details. In addition, the quantitative results on the YRE dataset are summarized in Table VIII. The proposed method achieves lower D_λ and D_s values compared with most competing approaches, indicating reduced spectral and spatial distortions in the fused images. It also attains the highest QNR value, suggesting a more balanced preservation of spectral information and spatial structures. These quantitative results are consistent with the visual comparisons in Fig. 10, where the proposed method shows

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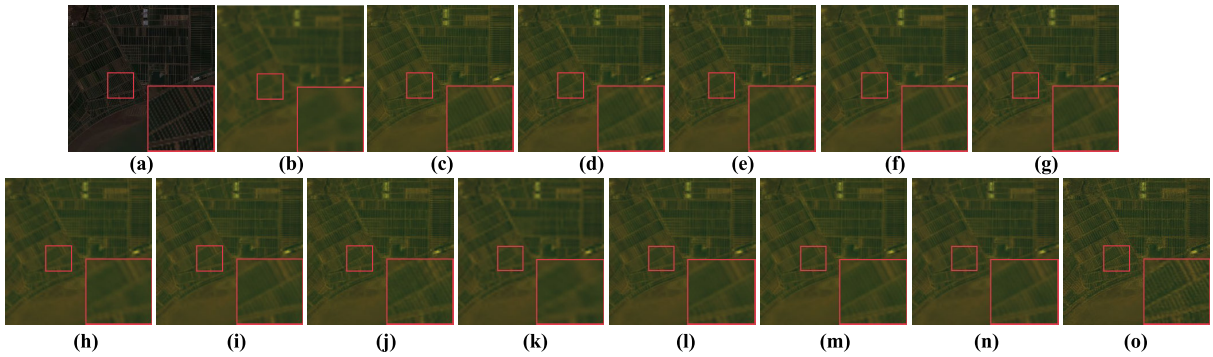


Fig. 10. Qualitative comparison of different methods on the YRE dataset. The results are shown by using the pseudocolor representation (R-60, G-29, and B-7). (a) MSI (Sentinel-2A). (b) HSI (Gaofen-5). (c) SSRNet. (d) ResTFNet. (e) MoG-DCN. (f) Fusformer. (g) DSPNet. (h) 3DT-Net. (i) DCTransformer. (j) PSTUN. (k) CYformer. (l) DMSN. (m) FusGAT. (n) DTUML. (o) Ours.

TABLE VIII
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE YRE DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Metric	SSRNet	ResTFNet	MoG-DCN	Fusformer	DSPNet	3DT-Net	DCTransformer	PSTUN	CYformer	DMSN	FusGAT	DTUML	Ours
$D_\lambda \downarrow$	0.0483	0.0471	0.0429	0.0490	0.0447	0.0398	0.0407	0.0396	0.0425	0.0406	0.0391	0.0406	**0.0371**
$D_s \downarrow$	0.1891	0.1788	0.1849	0.1757	0.1821	0.2037	0.1808	0.2098	0.1898	0.1889	0.1759	0.1788	**0.1741**
QNR $\uparrow$	0.7717	0.7825	0.7801	0.7839	0.7813	0.7646	0.7859	0.7589	0.7758	0.7782	0.7919	0.7879	**0.7983**

clearer spatial details and more faithful texture restoration. The agreement between quantitative metrics and qualitative observations further supports the effectiveness and robustness of the proposed method on real-world data.

V. CONCLUSION

In this article, we propose the RAMoE network for the fusion of LR-HSI and HR-MSI, aimed at effectively integrating spectral and spatial information while enhancing feature representation and fusion robustness. The ASM facilitates the sharing of attention information between HSI and MSI, thereby enabling seamless cross-modal interaction. In RAMoE, the RAEs attend to different regions according to their learned preferences, and their outputs are adaptively aggregated through a gating mechanism, which promotes complementary feature integration and improves the generalization capability of the model. In contrast, the IWE employs a spatially uniform and fixed weighting strategy, allowing it to capture globally consistent texture and structural information. Finally, the SSRM is introduced to get residual information from the original LR-HSI and HR-MSI. This module effectively restores fine-grained details by supplementing the fusion process with both spectral and spatial information. Extensive experiments on several datasets validate the effectiveness and robustness of the proposed network, demonstrating superior performance compared to state-of-the-art methods.

The fusion of LR-HSI and HR-MSI images to enhance image quality is often typically intended to improve the performance of more fine-grained remote sensing tasks. In real-world applications, remote sensing scenarios frequently cover large-scale areas, which imposes high demands on computational efficiency. For future research, we aim to develop a method capable of fusing large-scale HSI and MSI quickly while simultaneously enhancing the performance of various remote sensing tasks.

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